TAMILNADU SKILL DEVELOPMENT CORPORATION

Naan Mudhalvan Niral Thiruvizha 3.0 (2025-2026)

Review - 2

1. **Name of the College:** St.Joseph's College Of Engineering
2. **College code:** 3123
3. **Theme:** Robotics/ Drone/ Industry 4.0
4. **Problem statement:** How can we design an economical device, such as robotic boats, to efficiently collect plastic waste in marine ecosystems and integrate anti-pollutant technologies to protect and sustain the marine environment?
5. **Faculty Guide Details:** Mr. K. Pravinkumar M.E., (Ph.D.),

Assistant Professor, Department of Mechanical Engineering.

Team Leader Name	AU Reg.No	Year	Department	Mobile No
1. MONICA S	312323114083	III	Mechanical	8825634733
Team members				
1. NANDHAKUMAR G	312323114087	III	Mechanical	9344161321
2. MOHAN RAM S	312323114082	III	Mechanical	9003104317
3. KARTHIKEYAN G	312323114067	III	Mechanical	7550184432

6. Attach photos of the prototype/App/Software:

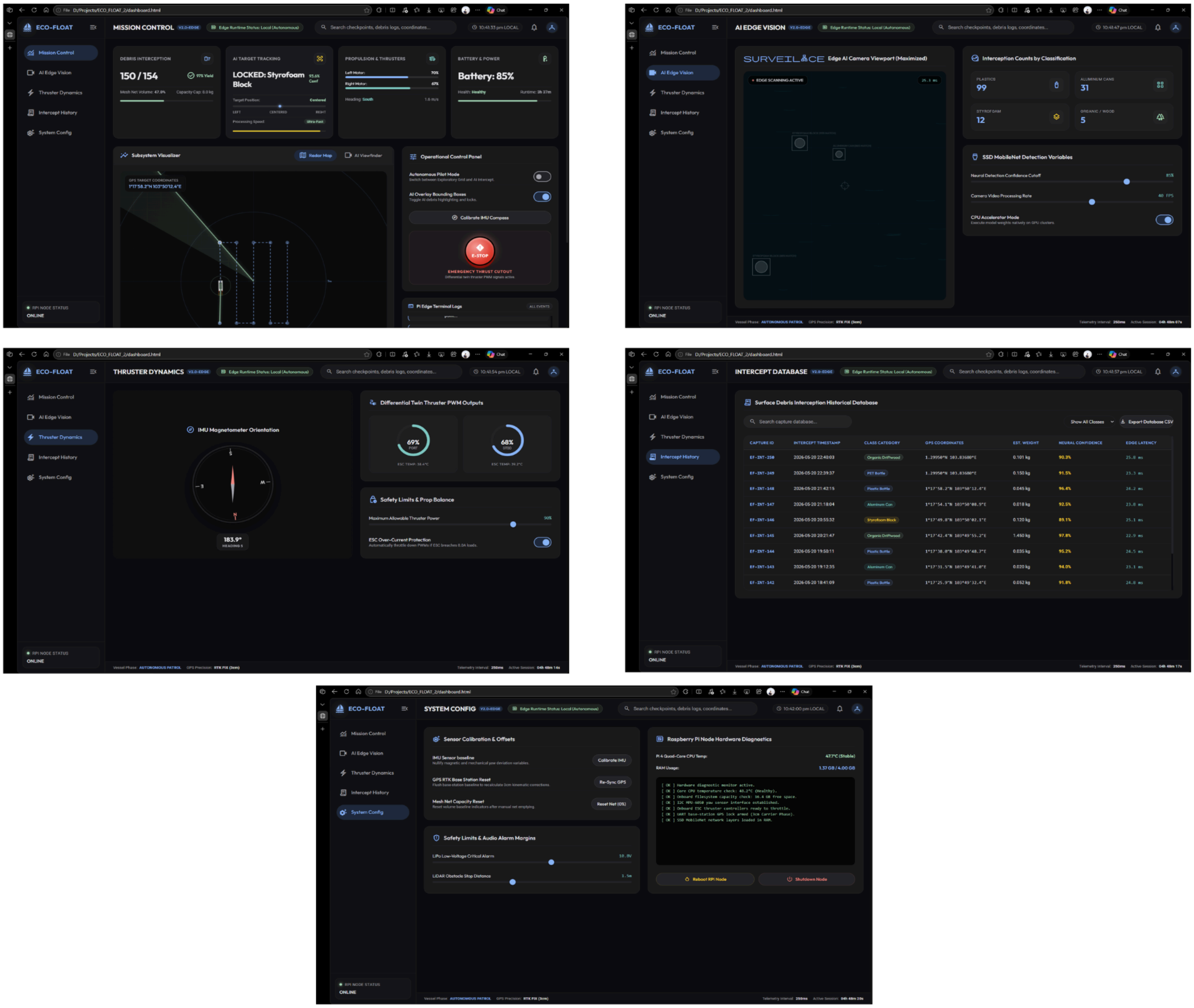

Fig. 1: Web dashboard

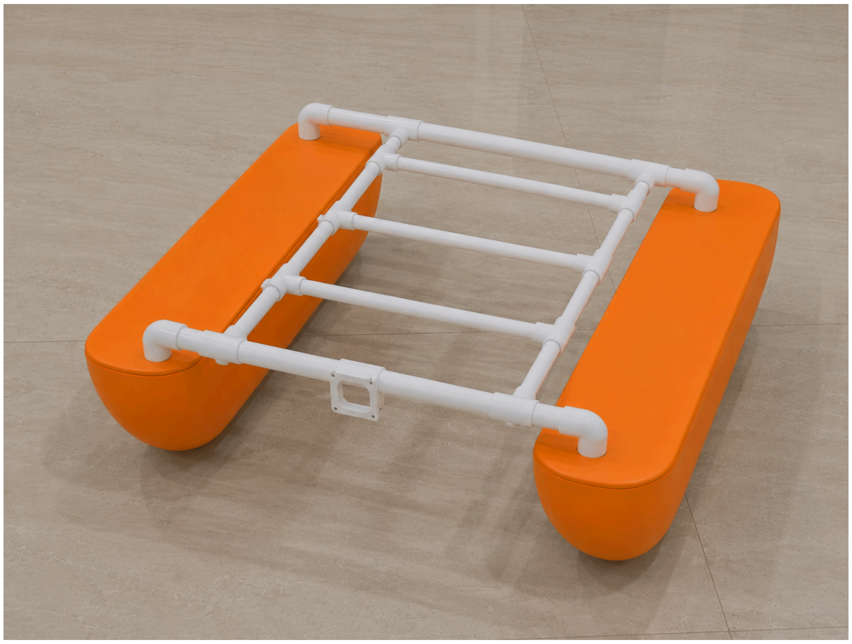

Fig. 2: Hull fabrication

7. Attach photos of the 2nd review being conducted:

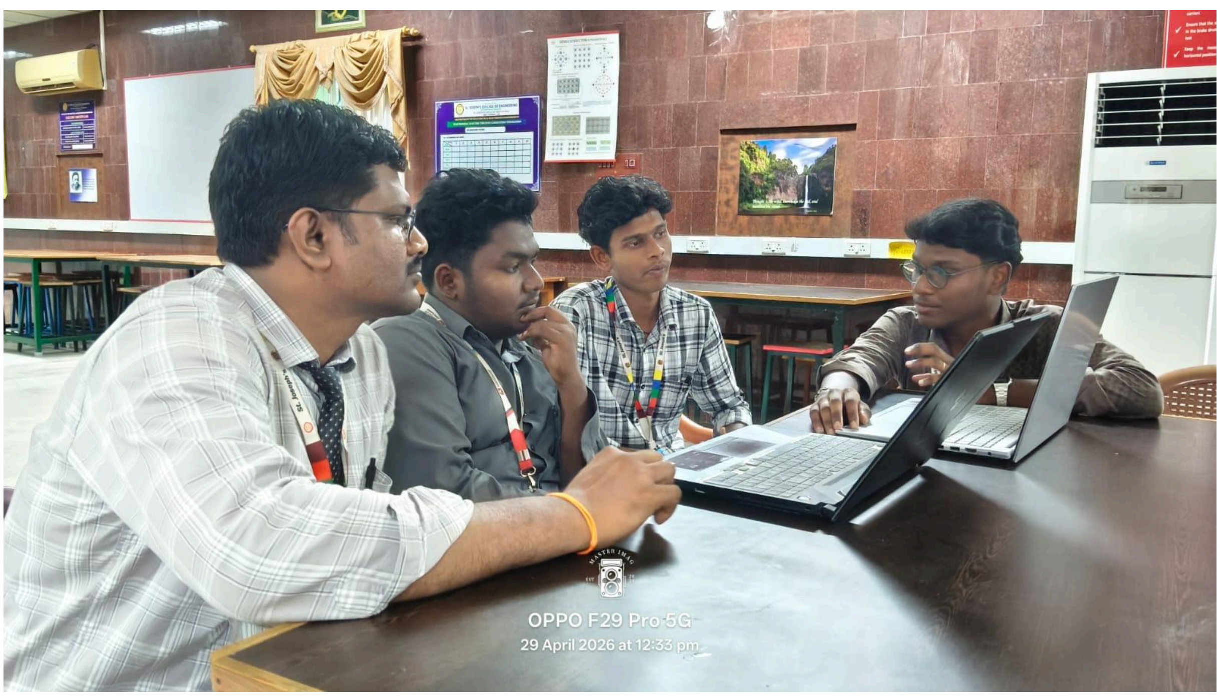

Fig. 3: Review 2 with Industry mentor and faculty guide

8. Detailed Description of the Project:

ECOFLOAT is an autonomous, AI-driven catamaran robotic system engineered for the active remediation and real-time environmental monitoring of stagnant urban water bodies. Built upon a hydrodynamically stable twin-hull catamaran chassis, the platform is specifically designed to address the escalating crisis of surface pollution through continuous, low-intervention operation. The core of its physical remediation capability lies in a passive, funnel-based intake mechanism that efficiently channels floating macro-debris into a high-capacity, rear-mounted collection net. Transitioning from its original conceptualization as a purely observational floating hyperspectral radiometer platform initially focused on detecting microplastics and measuring chlorophyll concentrations, this second iteration heavily emphasizes active, intelligent waste removal. At the heart of ECOFLOAT's autonomy is a sophisticated sensory and software suite driven by a front-mounted, high-resolution camera array. Utilizing advanced computer vision algorithms, the onboard AI processes live visual feeds to autonomously identify, classify, and track various types of floating debris. Instead of relying on manual teleoperation, the system employs systematic lawnmower pathing algorithms to ensure comprehensive, efficient, and complete coverage of designated aquatic zones. Furthermore, as the vessel navigates and collects waste, it simultaneously logs precise coordinate data to generate real-time pollution heatmaps. This dual-purpose approach not only physically extracts harmful solid waste to prevent ecological degradation but also transforms qualitative visual observations into quantitative, actionable data regarding debris concentration and distribution. Ultimately, ECOFLOAT provides a scalable, intelligent, and highly effective blueprint for restoring aquatic health and empowering environmental stakeholders with critical insights for targeted cleanup initiatives, ensuring cleaner, safer urban waterways for everyone.

UNDERTAKING

1. The college will provide the basic infrastructure and other required facilities to the students for the timely completion of their projects.
2. The college will undertake the financial and other management responsibilities of the project; the college will be responsible for following the timelines.
3. The college will ensure that the funds provided are utilized only for the purpose provided, and any remaining amount will be returned to the University after the completion of the project.
4. I hereby declare that the details furnished above are true and correct.

Signature of Faculty Guide **Signature of Principal**